WA Landmark Classification

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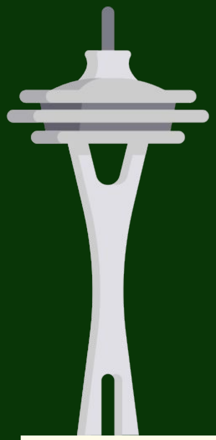
Photo by Robert Ritchie on Unsplash

Project Background



The goal of our project is to create a website that can identify **landmarks in Washington state (given an image).**

Tool Project

- Python
- Streamlit

Image Classification

- TensorFlow
- Transfer Learning

Previous Work

- Landmark classification Kaggle challenge
- How ours is different?

Data Sources

Google Landmarks Dataset v2

- 4+ million images → 200,000 locations
- Narrowed to landmarks in WA state - reduced to 296 locations
- Fields: id, name, wikimedia image url

Wikimedia

- Location
- Supercategory
- Latitude and Longitude

Personal Photos

- Space Needle
- Suzzallo Allen Library
- Drumheller Fountain
- Historic Chinatown Gate

Use Case 1 - Classify an Image

User: the Tourist

Uploads an image to be classified

↓

Model classifies image

↓

Interface shows the result, the accuracy, the landmark category, and location to the user

Use Case 2 - Search for a Landmark

User: the Tourist

Searches the name of the landmark

↓

Landmark classes are queried

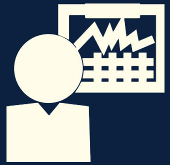
↓

Interface shows the user if the name is in the dataset and provides landmark details

Use Case 3 - Ask for a new landmark

Submits a feedback form with text and/or images

Email sent to model maintainer



Model maintainer uses package documentation to update model

User: the Model Maintainer

Component Design - Modular UI

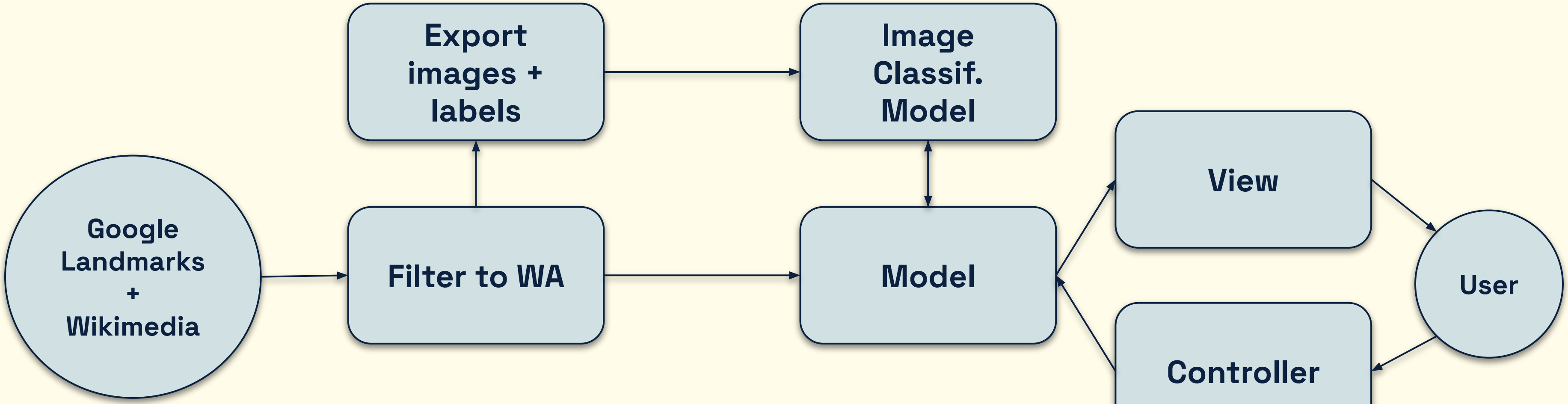
Home.py

Landing page

- Classifier.py
- Search_page.py
- Model_Analysis.py
- Feedback.py

Component Design - Backend

Pipe & Filter + MVC



Live Demo!

Photo by Thom Milkovic on Unsplash

Lessons Learned

Modularization of components	Organized code Easy collaboration
Streamlit can be inflexible	Multipage app naming incompatible w pylint AppTest limitations
Transfer learning rocks	Without transfer learning: ~30% accuracy With transfer learning: 87% accuracy
PyTorch vs. TensorFlow	Switched to PyTorch after Technology Review Final product uses TensorFlow

Future Work

WA State GeoGuessr

- Fun way to engage users!

Add more WA landmarks

- Requires manual data collection or merge of existing datasets

Expand landmarks outside WA

- Requires more compute resources

Thank You!